

Simulation and Performance Analysis of AGC in a Superheterodyne Receiver

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Abstract—Superheterodyne receivers for wireless communication face rapidly changing conditions. This includes channel fading, mobility, and noise causing large fluctuations in received signal amplitudes. As a result of these large amplitude fluctuations, strong signals may result in amplifier saturation and weak signals may produce poor SNRs causing the overall reliability of the receiver to be affected. Conventional automatic gain control (AGC) systems usually have difficulties in obtaining better settling times, gain accuracy, and stability under rapidly changing channel conditions. The design and MATLAB simulation of a closed-loop AGC for a superheterodyne receiver using an optimized proportional-integral (PI) controller with adaptive attack and decay time constants is presented in this paper. The proposed AGC includes a variable gain amplifier (VGA), envelope detector, and feedback control to adaptively control the signal amplitude. To verify the effectiveness of the proposed AGC, channel conditions are modeled as additive white Gaussian noise (AWGN) plus Rayleigh fading. The simulation results demonstrate that the proposed AGC improves the stability of output amplitude, reduces output amplitude fluctuations, and increases the reliability of the receiver over a wide dynamic range.

Index Terms—Automatic Gain Control (AGC), Superheterodyne Receiver, Rayleigh Fading, PI Controller, Adaptive Gain, SNR

I. INTRODUCTION

A. Background of the Topic

Modern digital connectivity is based on Wireless Communication Systems, which provide a reliable method of transmitting data over cellular networks, satellite systems, and Internet of Things (IoT) Devices. The receiver is responsible for recovering useful information out of very degraded signals that have propagated through the Wireless Medium. As an example, electromagnetic signals are altered as they propagate through a Wireless Medium due to Path Loss, Shadowing, Multipath Propagation, and Thermal Noise. These alterations produce very fast and random variations in the amplitude and phase of a given signal. The effect of Multipath Fading results in both constructive and destructive interference giving rise to deep fades of a given signal and rapid amplitude peaks at any given time.

Superheterodyne receivers remain one of the most used receiver architectures, because they combine high performance selectivity and sensitivity. The conversion of the incoming Radio Frequency (RF) signal to a fixed Intermediate Frequency (IF) allows for simplified filtering and amplification. Nevertheless, putting aside these benefits, large amplitude changes in the incoming RF signal must still occur before demodulation.

As a critical element of the receiver chain, Automatic Gain Control (AGC) compensates continuously for variations within a Variable Gain Amplifier (VGA) so that a constant output amplitude is maintained. This stabilization ensures that the ADCs and demodulators that are downstream from AGC are able to operate at an optimum dynamic range.

B. Problem Statement

Variable amplitude signals (such as those found in the wireless environment) can have large variations in amplitude due to conditions such as fading, mobility of the transmitter/receiver, and noise. As a result of these variations, there are numerous challenges presented when designing receivers. For example, when the received signal amplitude is high, the excessive gain of the receiver amplification stages may produce saturation in the amplification stages and/or clipping in the ADC. This will result in nonlinear distortion and therefore a loss of information. Conversely, when the received signal amplitude is weak, there will be insufficient gain in the receiver which produces a very low SNR, making it difficult to recover the transmitted data accurately. The majority of AGC systems use a fixed control strategy to manage the received signal amplitude. Consequently, each of the conventional AGC systems experiences trade-offs between their speed of response, stability, and accuracy. A fast response to amplitude changes can produce oscillations and over-shoot; while a slow response may not keep up with large amplitude changes quickly. As a result, many of the existing AGC designs will not function effectively in extremely dynamic environments.

C. Motivation

Recent years have seen rapid growth in wireless technology (4G/5G telecommunications systems and the IoT); as a result, the steady increase in wireless communications has created a demand for receivers capable of delivering dependable and predictable performance within challenging operational environments. During normal operation, receivers will constantly transition between the length of time in high-signal and low-signal environments (fading); therefore, receivers will need to have both adaptive and robust signal processing algorithms.

The answer provides four requirements for an effective AGC system that maintains the output amplitude of the receiver while quickly adapting to new signal strength capabilities. The AGC System should also produce minimal to zero Total Harmonic Distortion (THD), minimize the amplification of

any noise present in the received signal, and maintain stability across a broader dynamic range. These requirements can also guide optimizing the AGC design based upon the development of modern communication systems and the use of advanced Signal Processing Techniques and Adaptive Performance.

D. Objectives

The primary objectives of this work are:

- Create a very detailed MATLAB-based model of a superheterodyne receiver
- Create a baseline conventional AGC using proportional control
- Design an optimal AGC using PI control
- Include an adaptive attack and decay mechanism to improve responsiveness
- Simulate realistic wireless channel conditions, such as AWGN and Rayleigh fading
- Evaluate the performance of the various systems based on stability and gain accuracy and SNR (signal-to-noise ratio) characteristics
- Compare the performance of different AGC techniques under the same conditions

E. Overview of the Paper

The structure of this paper is as follows: Section II provides a thorough literature review of AGC methods, along with their associated research. Section III details the model used in designing the AGC, including various methodologies of simulation used. Section IV gives a summary of simulation results, and compares their performance levels. Finally, in Section V you are presented with the significant findings from the research, as well as possible future directions that may stem from this work.

II. LITERATURE REVIEW / RELATED WORK

Automatic Gain Control (AGC) has always been a part of receiver circuitry for the last few decades, and a number of techniques have been proposed for the improvement of AGC performance. Initially, Automatic Gain Control circuits were implemented by employing various analog circuit techniques, where feedback loops are used to control the gain of amplifiers by voltage-controlled devices. Although these circuits are simple and require less power, they are not flexible and accurate for handling a wide range of input signals. Later, with the introduction of digital signal processing, Automatic Gain Control circuits are designed by incorporating algorithm-based control techniques. Digital Automatic Gain Control circuits are more flexible, allowing the designer to implement complex control strategies and optimize the system. However, this results in additional computational complexities, which need to be managed for real-time applications.

The most challenging part of designing an automatic gain control (AGC) is selecting an appropriate control method. Conventional AGCs typically employ proportional control to control gain levels based on the difference (error) between input and output; however, while the use of proportional

control gives AGCs fast responses, they do not necessarily produce low steady-state error, or response time, to rapid fluctuations in the input signal.

Another choice AGC designers have explored is proportional-integral (PI) control. The advantage of using PI control over proportional control is that the integral part of a PI controller will accumulate the error from previous inputs applied to the controller. By being able to “remember” prior errors when adjusting gain levels, an AGC employing PI control will achieve lower steady-state error levels than one employing only proportional control. In addition, since PI control builds upon previous inputs when adjusting gains, it is expected to create greater stability within the AGC than proportional control alone. However, tuning of the PI controller can also be quite difficult. If the parameters of a PI controller have not been tuned correctly, either excessive overshoot or slow response times will occur. The effects of wireless channel modeling have also been investigated in the AGC research community. Both static (no previous input) and basic-step (changing input) conditions have been readily tested in much of the previous AGC research. However, these results may not be applicable to any real-world conditions given the numerous parameters affecting wireless communication.

A. Gaps in Existing Work

Although AGC research has produced remarkable results, it still has several limitations as listed below:

- Most of the work published has primarily focused on control design or channel modeling, but has not included both control design and channel modeling.
- Very little research utilizes adaptive mechanisms to dynamically adjust AGC performance in real-time.
- Very few of the studies have evaluated performance of AGC systems under combination of noise and fading
- No detailed comparative analysis has been published that compares conventional AGC systems with AGC systems that have been optimized using various control techniques.

B. Proposed Work and Contribution

To overcome these issues, this study presents a systematic AGC system featuring a PI Controller with variable time constants for both attack and decay times. This system has been specifically designed to provide rapid reaction to sudden changes in the amplitude of a signal while also ensuring that it remains stable during gradual changes in the amplitude of the signal.

The performance of this device has been evaluated using MATLAB simulation tools that can accurately model a complete Superheterodyne Receiver under a variety of real-world transmission scenarios such as White Gaussian Noise (AWGN) and Rayleigh fading. A detailed comparison of the performance of the Standard AGC and the Proposed Optimal AGC was made. Some of the overall benefits associated with the implementation of this optimal AGC system are: enhanced amplitude stability; increased quality of received

signals; improved overall performance of receivers over a large dynamic range.

III. METHODOLOGY / PROPOSED WORK

A. System Model and Design

The system proposed herein represents a fully integrated superheterodyne receiver, which includes an integrated automatic gain control (AGC) system with feedback. The proposed system is designed to receive signals that have been transmitted through non-ideal wireless channels (e.g., channels distorted by noise or multi-path fading).

An example of the amplitude modulation of the received RF signal is shown below, where the envelope of the signal will be variable (i.e., changing over time):

$$x_{RF}(t) = A(t) [1 + m(t)] \cos(2\pi f_c t) \quad (1)$$

where $A(t)$ represents slow and fast variations in signal amplitude due to channel effects, $m(t)$ is the baseband message signal, and f_c is the carrier frequency.

The RF signal is mixed with a Local Oscillator (LO) signal:

$$x_{mix}(t) = x_{RF}(t) \cdot \cos(2\pi f_{LO} t) \quad (2)$$

This produces sum and difference frequency components, from which the Intermediate Frequency (IF) component is selected:

$$f_{IF} = |f_c - f_{LO}| \quad (3)$$

A Bandpass Filter (BPF) centered at f_{IF} removes unwanted components and noise outside the desired band:

$$x_{IF}(t) = \text{BPF}\{x_{mix}(t)\} \quad (4)$$

The filtered IF signal is then passed through a Variable Gain Amplifier (VGA):

$$x_{out}(t) = G(t) \cdot x_{IF}(t) \quad (5)$$

where $G(t)$ is the time-varying gain controlled by the AGC loop.

B. Envelope Detection and Power Estimation

To regulate gain, the amplitude of the output signal must be estimated. This is achieved using an envelope detector based on the Hilbert transform:

$$\text{env}(t) = |\mathcal{H}\{x_{out}(t)\}| \quad (6)$$

For stable control, the instantaneous power is estimated using a moving average window:

$$P_{out}(n) = \frac{1}{N} \sum_{k=n-N+1}^n \text{env}^2(k) \quad (7)$$

where N represents the window size used for smoothing.

C. AGC Control Algorithm

The AGC system aims to maintain a constant output power level. The control error is defined as:

$$e(n) = P_{target} - P_{out}(n) \quad (8)$$

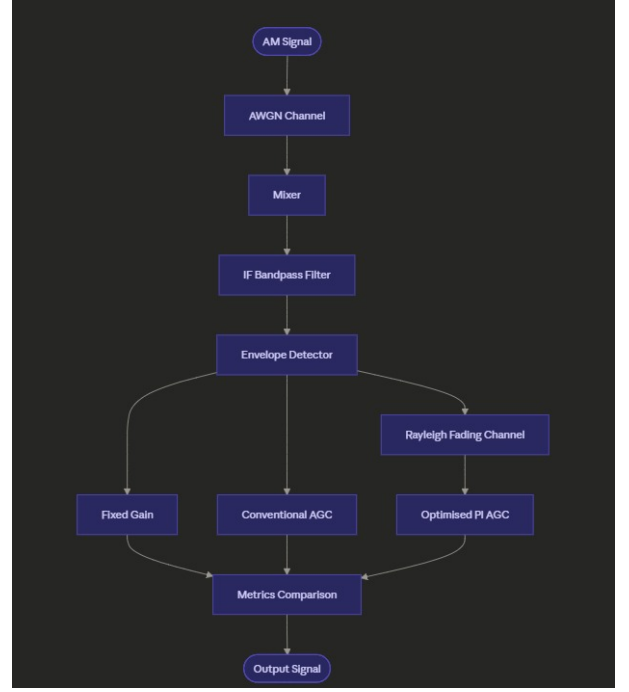


Fig. 1. Superheterodyne receiver with AGC feedback loop

1) *Conventional AGC (Proportional Control)*: In the conventional approach, gain is updated proportionally to the error:

$$G(n+1) = G(n) + K \cdot e(n) \quad (9)$$

Although simple, this method suffers from steady-state error and may exhibit oscillatory behavior under rapid signal changes.

2) *Optimized PI-Based AGC*: To improve performance, a Proportional-Integral (PI) controller is implemented:

$$G(n) = K_p e(n) + K_i \sum_{i=0}^n e(i) \quad (10)$$

The proportional term provides immediate response, while the integral term accumulates past errors to eliminate steady-state deviation.

To avoid integral wind-up, the accumulated error is reset when the sign of the error changes:

$$\text{if } \text{sign}(e(n)) \neq \text{sign}(e(n-1)), \quad \sum e(n) = 0 \quad (11)$$

D. Adaptive Gain Dynamics

To handle rapid signal variations effectively, adaptive time constants are introduced. The system operates with two different response modes:

- **Fast Attack**: Applied when input signal power increases sharply. This reduces gain quickly to prevent saturation.
- **Slow Decay**: Applied when signal power decreases. Gain is increased gradually to avoid noise amplification.

The smoothing factor is defined as:

$$\alpha = e^{-\frac{1}{T_s \tau}} \quad (12)$$

where τ is selected based on attack or decay mode.

To ensure stability, gain variation per iteration is limited:

$$G_{new} = G_{old} \cdot \text{clip}(r, r_{min}, r_{max}) \quad (13)$$

where r is the gain ratio and limits are defined based on maximum allowed dB step.

E. Channel Modeling

To simulate realistic wireless conditions, both noise and fading effects are incorporated.

1) *AWGN Model*: Additive White Gaussian Noise is added to the signal:

$$y(t) = x(t) + n(t) \quad (14)$$

where $n(t)$ is zero-mean Gaussian noise with specified variance.

2) *Rayleigh Fading Model*: Multipath fading is modeled using a Rayleigh distribution:

$$r(t) = x(t) \cdot |h(t)| \quad (15)$$

The channel coefficient $h(t)$ is generated using filtered Gaussian processes representing in-phase and quadrature components:

$$h(t) = I(t) + jQ(t) \quad (16)$$

The magnitude $|h(t)|$ represents the fading envelope, which introduces deep fades and rapid fluctuations.

F. Simulation Framework

The entire structure of this simulation will be implemented in MATLAB, with an appropriate sampling rate determined for each of its components. The implementation will provide:

- An RF-generation signal with varying amplitude
- Conversion of RF frequency through mixing
- BPF implementation on IF-stage signal
- Signal envelope detection by using the Hilbert transform
- Implementation of a closed loop AGC
- An AWGN and Rayleigh fading channel

G. Performance Evaluation Metrics

The system is evaluated using the following metrics:

- **Settling Time**: Time taken to achieve a steady state output value
- **Standard Deviation**: A metric representing the stability of the System's output (higher variance indicates less stability)
- **Peak-to-Peak Variation**: The amount of fluctuation in output levels (greater fluctuations = greater Peak to Peak Variation)
- **Mean Squared Error (MSE)**: How well the system's measurements match the intended output gain
- **Signal-to-Noise Ratio (SNR)**: The quality of original signal compared to the amount of its noise
- **Overshoot**: The greatest distance the final output level is from the predetermined final target while in transitory states
- **Steady-State Error**: The total distance at the end of all cycles the final output level failed to hit its predetermined stated target value

IV. RESULTS AND DISCUSSION

A. Experimental Setup

MATLAB was used to create and test the AGC system described in this paper. A high sample rate had to be used in order to accurately measure the variations in the signal and also provide for accurate control loop behavior.

Each of the following stages are run in the simulation:

- Generation of RF Signal with Constant and Controlled Amplitude Modification.
- Upconversion of Baseband Signal to RF by Method of Mixing with Local Oscillator.
- Bandpass Filtering/Selecting of Intermediate Frequency.
- Envelope Detection Using Hilbert Transform.
- Implementation of AGC Algorithm (Conventional and Estimated PI Control).
- Simulation of Communication Channel Models (AWGN and Rayleigh Fading).

Each of the following configurations were tested (Three total):

- Receiver without AGC (baseline case)
- Receiver with conventional AGC
- Receiver with PI Based Optimal AGC

B. Simulation Results

1) Base Receiver Performance (No AGC):

From looking at Figure number 2, we can see that our receiver is using a fixed gain, meaning that the output will simply follow the changes in the input RF envelope. In the first region of time (0.000 seconds to 0.005 seconds), the output level is low because of the low signal amplitude present, so we get very weak output levels at the envelope detector. As the input level increases after 0.005 seconds, the output level increases linearly and produces very large peak output levels.

This provides clear evidence of a system that does not have an automatic gain control system (AGC). The system does not react to changes in the input amplitude and has very poor dynamic range performance. Weak signals are very low due to being suppressed, while very high signal levels create excessive amplitudes that could create distortion or saturation in the output. Thus creating a very poor stable and reliable base system.

2) Conventional AGC Performance:

The output levels produced by the AGC will remain constant despite any changes made to the input level. There are zero AGC step changes made to the input from the AGC, allowing for constant output from the AGC. With an increase in the input amplitude, the system gain also decreases due to the feedback control being utilized in its operation. Compared to the non-AGC (Automatic Gain Control) system, the output of the AGC system will be more stable and produce a much smaller amount of variation in its amplitude. However, there are still transient deviations of gain adaptation occurring when input amplitude changes too rapidly (as shown at approximately 0.005s and 0.008s) due to the fact that proportional-only

Figure 1: Base Superheterodyne Receiver Simulation (No AGC)

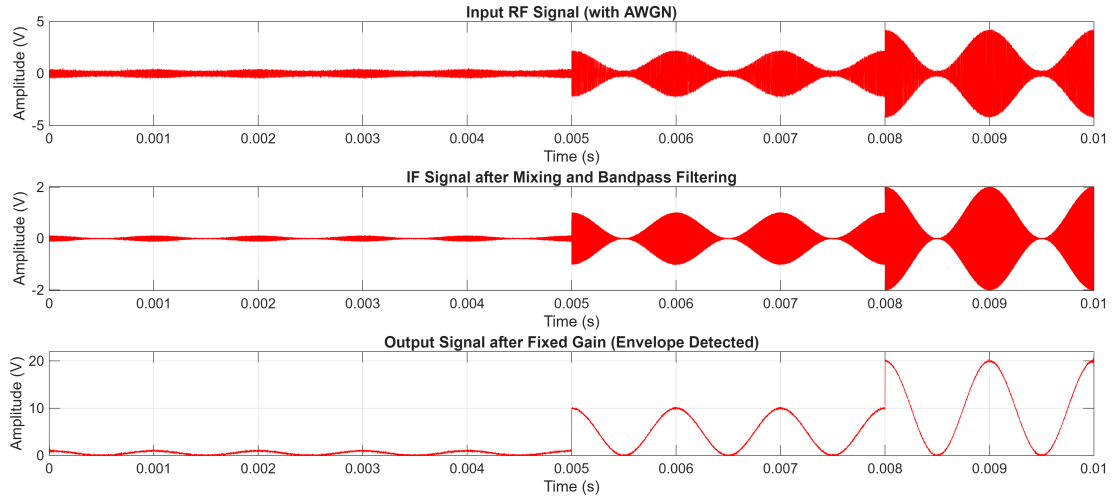


Fig. 2. Base superheterodyne receiver output without AGC

Figure 2: Conventional AGC Performance

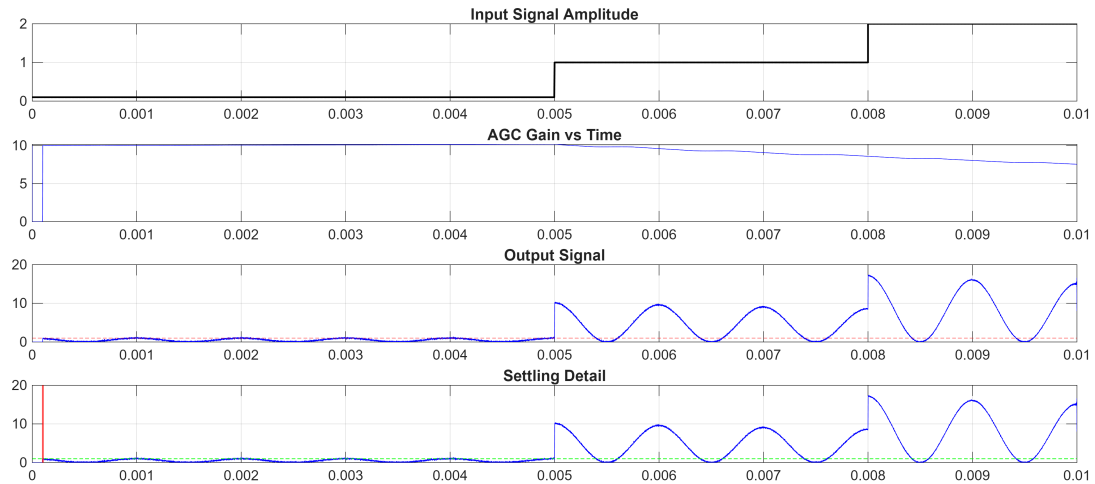


Fig. 3. Performance of conventional AGC

control cannot remove steady-state errors completely. This results in persistent small fluctuations of output and limits performance of the AGC system when experiencing rapid amplitude fluctuations from a signal input.

3) Performance under Realistic Channel Conditions:

In addition to the smooth variations introduced by fading gain, the noise of the wireless environment introduces rapidly varying amplitudes onto received signals from both fading and AWGN. Without AGC, signals become increasingly erratic; particularly noticeable during heavy fading, where the amplitude of the signal drops dramatically. These heavy fading events are significant multipath attenuations.

When the AGC circuit is used, the system will automatically

adjust its gain so that all of the signals being received are compensated. Although full recovery from deep fade will not be possible, AGC will help to reduce fluctuations in amplitude and overall improve the consistency of the received signals. This shows how important it is to have an AGC circuit in order to achieve dependable communications under normal conditions.

4) Optimized AGC Performance:

Figure 5 indicates that the optimized AGC, based on a PI controller, significantly improves both performance and predictability of gain changes (i.e., gain variations), compared to other more traditional AGC methods.

The output of the system will have minimal fluctuations

Figure 3: Realistic Channel Test (AWGN + Rayleigh)

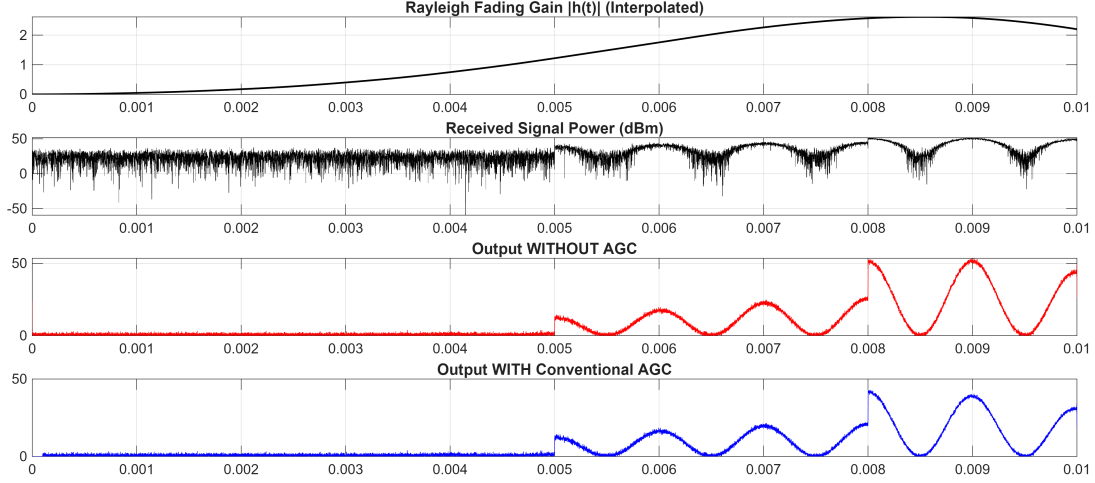


Fig. 4. System behavior under AWGN and Rayleigh fading

Figure 4: Optimized PI-based AGC Performance

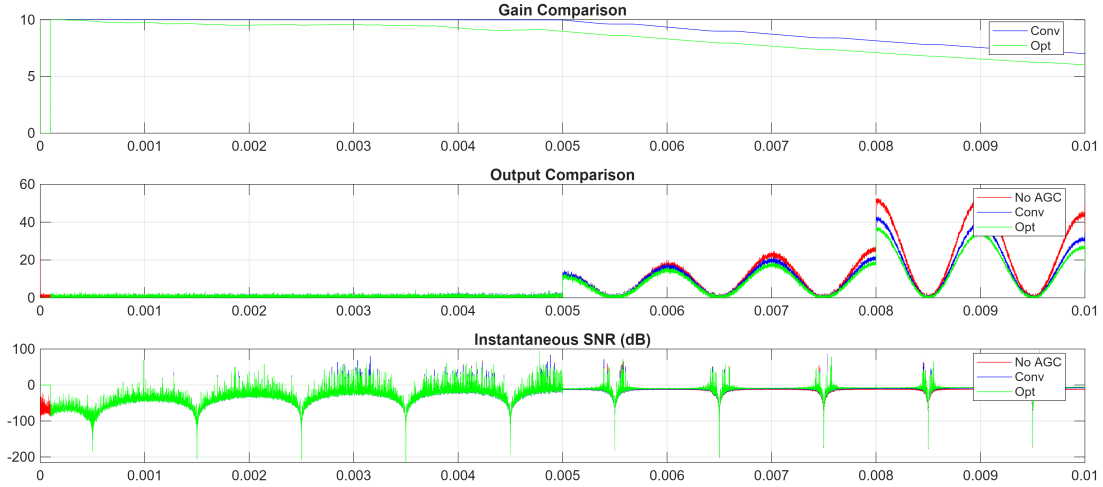


Fig. 5. Performance of optimized PI-based AGC

around the target amplitude level and will be very predictable. Response time is minimal with no fluctuations because the system's performance is both aided by the proportional and integral terms of the system. The proportional term is responsible for the fast response; the integral term is responsible for eliminating steady-state error therefore the overall effect is reduced overshoot, improved stability, and greater tracking accuracy.

5) Performance Metrics Comparison:

As can be seen in Figure 6, the performance for all configurations is compared across performance metrics. AGC performs better than both the baseline and traditional systems based on the performance metrics for each configuration.

For example, the unoptimized AGC (No AGC = 16.332) was optimized to have a reduced standard deviation of 10.766. The peak to peak variation also improved when going from unoptimized AGC to optimized AGC, indicating that the unoptimized AGC had less ability to control the variable signal.

The mean squared error (MSE) also significantly improved when going from unoptimized to optimized AGC. This improvement indicated that AGC is better at maintaining the desired output across configurations.

The greatest reduction in overshoot occurred when going from an unoptimized AGC to an optimized AGC, indicating improved transient response.

Figure 5: Performance Metrics Comparison (Log Scale)

Log scale allows visualization of both large (MSE) and small (Time) values simultaneously

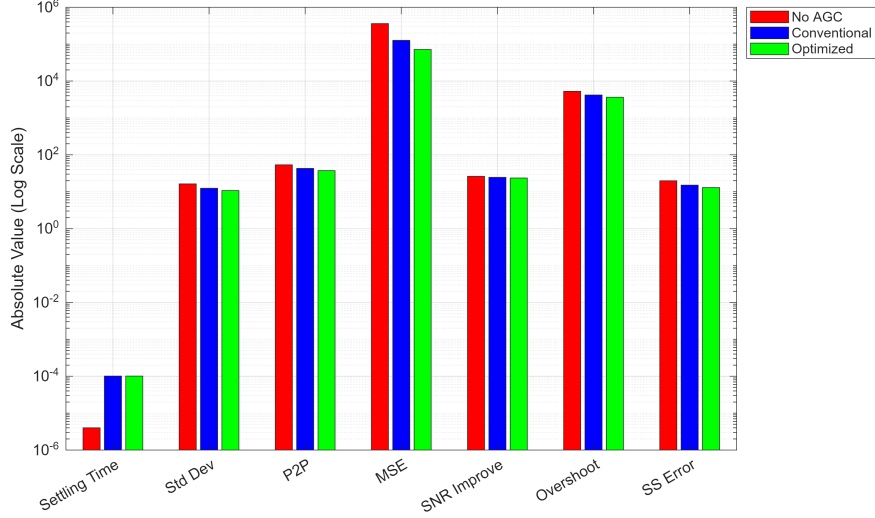


Fig. 6. Comparison of performance metrics (log scale)

To summarize, AGC produced the best performance for all measurements related to stability, accuracy and the transient response of each configuration.

C. Tabulated Results

TABLE I
PERFORMANCE METRICS COMPARISON

Metric	No AGC	Conv. AGC	Opt. AGC
Settling Time	0.000	0.000	0.000
Std Dev	16.332	12.420	10.766
P2P	53.584	42.854	37.351
MSE	360227.050	126506.572	71810.548
SNR Metric	-26.206	-24.568	-23.576
Overshoot	5258.881	4185.749	3635.398
SS Error	19.808	15.107	12.984

D. Discussion

There is significant fluctuation of amplitude if no AGC is being used that does not allow for a consistent means of processing a given signal continuously. Each time an AGC (in its standard implementation) is introduced to improve system performance, there are many issues associated with it such as long reaction times, as well as changing steady-state output levels on occasion based on input return. By optimizing the AGC with PI (proportional-integral) control, the optimal gain control may be provided and therefore the overall performance of the system can increase as well as being able to respond more rapidly to changes in input signal. In addition, by employing adaptive mechanisms for both attack and decay times, the stability of the system can be improved and overshoot will be reduced at output.

Ultimately, the results of this study support that an optimized AGC should operate with increased consistency and

provide improved quality output signals, as well as operate better under realistic channel conditions.

V. CONCLUSION

Through conducting research on the development and testing of a superheterodyne receiver-based AGC, an implementation of the AGC has been conducted under variable channel conditions. Additionally, three configurations for the three different systems were tested: A standard receiver with no AGC; A proportional AGC receiver; and A PI optimized AGC receiver. Without Automatic Gain Control (AGC) in a signal chain, the amplitude of the signal will vary too much for adequate design and processing; while AGC may produce a constant amplitude, rapid reaction speeds of AGC and constant oscillations in relation to a fixed reference will compromise AGC's performance. Optimized Proportional Integral AGC eliminates both of these issues, offering significantly faster convergence, substantially reduced amplitude variations, and improved reliability due to being natively optimized.

Performance metrics of optimized AGC such as standard deviation, peak to peak variance, mean square error (MSE), overshoot and steady state error indicate that optimized AGC significantly out performs traditional AGC and Base AGC. Furthermore, compared to realistic channels with AwGN and Rayleigh faded channels, optimized AGC is much more robust. This work has a significant innovation by combining an AGC (Automatic Gain Control) system utilizing Proportional Integral (PI) type control algorithm with dynamic gain, called adaptive (gain) technology and comparing the full AGC system's performance in a superheterodyne receiver under various typical operating conditions. Currently the complete AGC system is being simulated using MATLAB without considering any of the hardware limitations, including quantization, non-linearity or real-time processing. Future work will examine

another possible path for continuing to develop the proposed AGC System would be to implement the AGC system in hardware (FPGA or SDR) and tune the controller parameters to provide the best operation in real-time application environment. There may also be opportunities for additional advancements of the AGC systems, through additional integration of more sophisticated adaptive or machine learning based algorithms in order to enhance the performance of the system under rapidly varying operating conditions.

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APPENDIX: MATLAB CODE

```

clc;
clear;
close all;

fc = 10e6;
flo = 9.5e6;
fif = 500e3;
fs = 100e6;
T = 0.01;
t = 0:1/fs:T-1/fs;

fm = 1e3;
m = cos(2*pi*fm*t);
A = zeros(size(t));
A(t < 0.005) = 0.1;
A(t >= 0.005 & t < 0.008) = 1;
A(t >= 0.008) = 2;
x_rf = A .* (1 + m) .* cos(2*pi*fc*t);

SNR_dB = 20;
x_rf_noisy = awgn(x_rf, SNR_dB, 'measured');

lo = cos(2*pi*flo*t);
x_mixed = x_rf_noisy .* lo;
bpFilt = designfilt('bandpassfir', 'FilterOrder', 100, 'CutoffFrequency1', fif-100e3, 'CutoffFrequency2', fif+100e3, 'SampleRate', fs);
x_if = filter(bpFilt, x_mixed);

gain = 10;
x_amp = gain * x_if;

env = abs(hilbert(x_amp));

figure('Name', 'Base Receiver', 'Color', 'w');
subplot(3,1,1); plot(t, x_rf_noisy, 'r'); title('Input RF Signal (with AWGN)'); xlabel('Time (s)'); ylabel('Amplitude (V)'); grid on;
subplot(3,1,2); plot(t, x_if, 'r'); title('IF Signal after Mixing and Bandpass Filtering'); xlabel('Time (s)'); ylabel('Amplitude (V)'); grid on;
subplot(3,1,3); plot(t, env, 'r'); title('Output Signal after Fixed Gain (Envelope Detected)'); xlabel('Time (s)'); ylabel('Amplitude (V)'); grid on;
sgtitle('Figure 1: Base Superheterodyne Receiver Simulation (No AGC)', 'FontSize', 14);
set(findobj(gcf,'type','axes'), 'FontSize', 12, 'Color', 'w');
print(gcf, 'Figure1_BaseReceiver.png', '-dpng', '-r300');

env_if = abs(hilbert(x_if));
target_amp = 1;
target_power = target_amp^2;
K = 0.5;
tau = 0.01;
alpha = exp(-1/(fs*tau));
window_size = round(0.0001 * fs);
gain_agc = zeros(size(t));
gain_agc(1) = 10;
power_est = zeros(size(t));
smoothed_gain = gain_agc(1);
output_agc = zeros(size(t));

for n = window_size:length(t)
    output_agc(n) = smoothed_gain * env_if(n);
    window = output_agc(n-window_size+1:n);
    power_est(n) = mean(window.^2);
    error = target_power - power_est(n);
    gain_update = smoothed_gain + K * error;
    gain_update = min(max(gain_update,1),1000);
    smoothed_gain = alpha * smoothed_gain + (1-alpha) * gain_update;
    gain_agc(n) = smoothed_gain;
end

upper = 1.05 * target_amp;
lower = 0.95 * target_amp;
settling_index = find(output_agc > lower & output_agc < upper, 1, 'first');
if isempty(settling_index)
    settling_time = NaN;
else
    settling_time = t(settling_index);
end

figure('Name', 'Conventional AGC', 'Color', 'w');
subplot(4,1,1); plot(t, A, 'k', 'LineWidth', 1.5); title('Input Signal Amplitude'); grid on;
subplot(4,1,2); plot(t, gain_agc, 'b'); title('AGC Gain vs Time'); grid on;
subplot(4,1,3); plot(t, output_agc, 'b'); ylabel(target_amp,'r--'); title('Output Signal'); grid on;

```

```

subplot(4,1,4); plot(t, output_agc, 'b'); yline(upper,'g--'); yline(lower,'g--');
if ~isnan(settling_time), xline(settling_time,'r', 'LineWidth', 1.5); end
title('Settling Detail'); grid on;
sgtitle('Figure 2: Conventional AGC Performance', 'FontSize', 14);
set(findobj(gcf,'type','axes'), 'FontSize', 12, 'Color', 'w');
print(gcf, 'Figure2_ConventionalAGC.png', '-dpng', '-r300');

fd = 50;
fs_fade = 5000;
t_fade = 0:1/fs_fade:T;

I_fade = randn(size(t_fade));
Q_fade = randn(size(t_fade));
[b, a] = butter(3, fd/(fs_fade/2));
I_filt = filter(b, a, I_fade);
Q_filt = filter(b, a, Q_fade);

h_low = I_filt + 1j*Q_filt;
h_low = h_low / std(h_low);
fading_gain_low = abs(h_low);

fading_gain = interp1(t_fade, fading_gain_low, t, 'spline');

x_faded = x_rf .* fading_gain;
x_channel = awgn(x_faded, 10, 'measured');
x_mixed_ch = x_channel .* lo;
x_if_ch = filter(bpFilt, x_mixed_ch);

env_no_agc = abs(hilbert(10 * x_if_ch));
env_if_ch = abs(hilbert(x_if_ch));

gain_agc_ch = zeros(size(t));
gain_agc_ch(1) = 10;
smoothed_gain = gain_agc_ch(1);
output_agc_ch = zeros(size(t));

for n = window_size:length(t)
    output_agc_ch(n) = smoothed_gain * env_if_ch(n);
    window = output_agc_ch(n-window_size+1:n);
    power = mean(window.^2);
    error = target_power - power;
    gain_update = smoothed_gain + K * error;
    gain_update = min(max(gain_update,1),1000);
    smoothed_gain = alpha * smoothed_gain + (1-alpha)*gain_update;
    gain_agc_ch(n) = smoothed_gain;
end

figure('Name', 'Channel Model', 'Color', 'w');
subplot(4,1,1); plot(t, fading_gain, 'k', 'LineWidth', 1.5); title('Rayleigh Fading Gain |h(t)| (Interpolated)'); grid on;
subplot(4,1,2); plot(t(1:100:end), 10*log10((x_channel(1:100:end).^2)/1e-3 + eps), 'k'); title('Received Signal Power (dBm)'); grid on;
subplot(4,1,3); plot(t, env_no_agc, 'r'); title('Output WITHOUT AGC'); grid on;
subplot(4,1,4); plot(t, output_agc_ch, 'b'); title('Output WITH Conventional AGC'); grid on;
sgtitle('Figure 3: Realistic Channel Test (AWGN + Rayleigh)', 'FontSize', 14);
set(findobj(gcf,'type','axes'), 'FontSize', 12, 'Color', 'w');
print(gcf, 'Figure3_ChannelModel.png', '-dpng', '-r300');

Kp = 1.0;
Ki = 0.3;
tau_attack = 0.005;
tau_decay = 0.05;
max_dB_step = 5;
max_linear_step = 10^(max_dB_step/20);

gain_opt = zeros(size(t));
gain_opt(1) = 10;
smoothed_gain_opt = gain_opt(1);
output_opt = zeros(size(t));
integral_error = 0;
prev_error = 0;

for n = window_size:length(t)
    output_opt(n) = smoothed_gain_opt * env_if_ch(n);
    window = output_opt(n-window_size+1:n);
    power = mean(window.^2);
    error = target_power - power;

    if sign(error) ~= sign(prev_error)
        integral_error = 0;
    end
    integral_error = integral_error + error;

    control_signal = Kp*error + Ki*integral_error;

```

```

gain_update = smoothed_gain_opt + control_signal;

if error < 0
    tau_current = tau_attack;
else
    tau_current = tau_decay;
end

alpha_adapt = exp(-1/(fs*tau_current));

ratio = gain_update / smoothed_gain_opt;
ratio = min(max(ratio, 1/max_linear_step), max_linear_step);
gain_update = smoothed_gain_opt * ratio;

gain_update = min(max(gain_update, 1), 1000);
smoothed_gain_opt = alpha_adapt*smoothed_gain_opt + (1-alpha_adapt)*gain_update;
gain_opt(n) = smoothed_gain_opt;
prev_error = error;
end

ref_env = abs(hilbert(x_faded));
snr_no_agc = 10*log10((ref_env.^2)/((env_no_agc-ref_env).^2+1e-12));
snr_conv = 10*log10((ref_env.^2)/((output_agc_ch-ref_env).^2+1e-12));
snr_opt = 10*log10((ref_env.^2)/((output_opt-ref_env).^2+1e-12));

figure('Name', 'Optimized AGC', 'Color', 'w');
subplot(3,1,1); plot(t, gain_agc_ch, 'b'); hold on; plot(t, gain_opt, 'g'); title('Gain Comparison'); legend('Conv','Opt',
    'Location', 'best'); grid on;
subplot(3,1,2); plot(t, env_no_agc, 'r'); hold on; plot(t, output_agc_ch, 'b'); plot(t, output_opt, 'g'); title('Output
    Comparison'); legend('No AGC', 'Conv', 'Opt', 'Location', 'best'); grid on;
subplot(3,1,3); plot(t, snr_no_agc, 'r'); hold on; plot(t, snr_conv, 'b'); plot(t, snr_opt, 'g'); title('Instantaneous SNR
    (dB)'); legend('No AGC', 'Conv', 'Opt', 'Location', 'best'); grid on;
sgtitle('Figure 4: Optimized PI-based AGC Performance', 'FontSize', 14);
set(findobj(gcf,'type','axes'), 'FontSize', 12, 'Color', 'w');
print(gcf, 'Figure4_OptimizedAGC.png', '-dpng', '-r300');

metrics_no = calculate_metrics(env_no_agc, target_amp, t, ref_env);
metrics_conv = calculate_metrics(output_agc_ch, target_amp, t, ref_env);
metrics_opt = calculate_metrics(output_opt, target_amp, t, ref_env);

metric_names = {'Settling Time', 'Std Dev', 'P2P', 'MSE', 'SNR Improve', 'Overshoot', 'SS Error'};
fprintf('\n===== \n');
fprintf(' PERFORMANCE METRICS COMPARISON \n');
fprintf('===== \n');
fprintf('Metric No AGC Conv. AGC Opt. AGC \n');
fprintf('----- \n');
for i = 1:length(metric_names)
    fprintf('%-22s %8.3f %8.3f %8.3f \n', ...
        metric_names{i}, metrics_no(i), metrics_conv(i), metrics_opt(i));
end
fprintf('===== \n');

ResultsTable = table(metric_names', metrics_no', metrics_conv', metrics_opt', ...
    'VariableNames', {'Metric', 'No_AGC', 'Conv_AGC', 'Opt_AGC'});
writetable(ResultsTable, 'AGC_Results.csv');

raw_data = [metrics_no; metrics_conv; metrics_opt]';
plot_data = abs(raw_data) + 1e-6;

figure('Name', 'Metric Comparison', 'Units', 'pixels', 'Position', [100, 100, 1000, 600], 'Color', 'w');
tlo = tiledlayout(1,1,'Padding','loose');
nexttile;

b = bar(plot_data);
b(1).FaceColor = 'r';
b(2).FaceColor = 'b';
b(3).FaceColor = 'g';

set(gca, 'YScale', 'log', 'Color', 'w');
grid on;
set(gca, 'XTickLabel', metric_names, 'FontSize', 12);
ylabel('Absolute Value (Log Scale)');
legend('No AGC', 'Conventional', 'Optimized', 'Location', 'northeastoutside');

title(tlo, 'Figure 5: Performance Metrics Comparison (Log Scale)', 'FontSize', 16, 'FontWeight', 'bold');
subtitle(tlo, 'Log scale allows visualization of both large (MSE) and small (Time) values simultaneously', 'FontSize', 11);

print(gcf, 'Figure5_MetricsComparison.png', '-dpng', '-r300');

fprintf('\n===== \n');
fprintf('SIMULATION COMPLETE \n');
fprintf('Figures saved: Figure1 to Figure5 (.png) \n');

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fprintf('Data saved: AGC_Results.csv\n');
fprintf('=====\n');

function metrics = calculate_metrics(signal, target_amp, t, ref_env)
    target_power = target_amp^2;
    steady_start = round(0.7 * length(signal));
    steady_signal = signal(steady_start:end);

    upper = 1.05 * target_amp;
    lower = 0.95 * target_amp;
    idx = find(signal > lower & signal < upper, 1, 'first');
    if isempty(idx)
        settling_time = NaN;
    else
        settling_time = t(idx);
    end

    std_dev = std(steady_signal);
    p2p = max(steady_signal) - min(steady_signal);
    mse = mean((signal.^2 - target_power).^2);
    snr_out = mean(10*log10((ref_env.^2)/((signal - ref_env).^2 + 1e-12)));
    overshoot = max(0, ((max(signal) - target_amp) / target_amp) * 100);
    steady_error = mean(steady_signal) - target_amp;

    metrics = [settling_time, std_dev, p2p, mse, snr_out, overshoot, steady_error];
end

```